In [27]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df=pd.read_excel(r"C:\Users\Lenovo\Desktop\Python\Assignments\Financial_Sample.x
```

In [3]:
```
df.shape
```

Out[3]:
```
(700, 16)
```

In [4]:
```
df.head(5)
```

Out[4]:

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Di
0	Government	Canada	Carretera	NaN	1618.5	3	20.0	32370.0	
1	Government	Germany	Carretera	NaN	1321.0	3	20.0	26420.0	
2	Midmarket	France	Carretera	NaN	2178.0	3	15.0	32670.0	
3	Midmarket	Germany	Carretera	NaN	888.0	3	15.0	13320.0	
4	Midmarket	Mexico	Carretera	NaN	2470.0	3	15.0	37050.0	

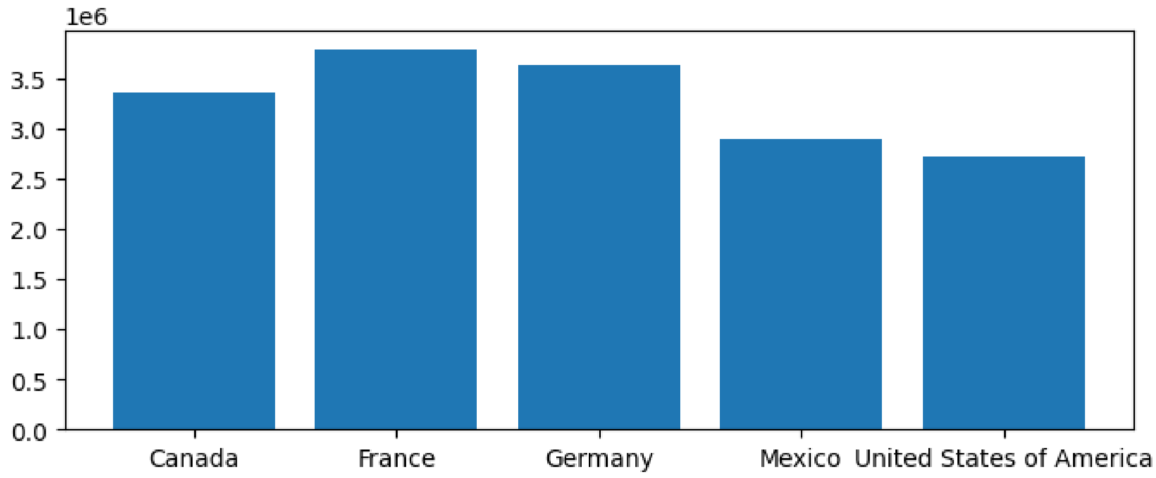
In [5]:
```
#1.Total profit by country using a bar chart
```

In [8]:
```
pbc=df.groupby('Country')['Profit'].sum().reset_index()
pbc
```

Out[8]:

	Country	Profit
0	Canada	3359437.935
1	France	3781020.780
2	Germany	3624167.730
3	Mexico	2889313.870
4	United States of America	2723336.745

In [10]:
```
plt.figure(figsize=(8,3))
plt.bar(pbc['Country'],pbc['Profit'])
plt.show()
```



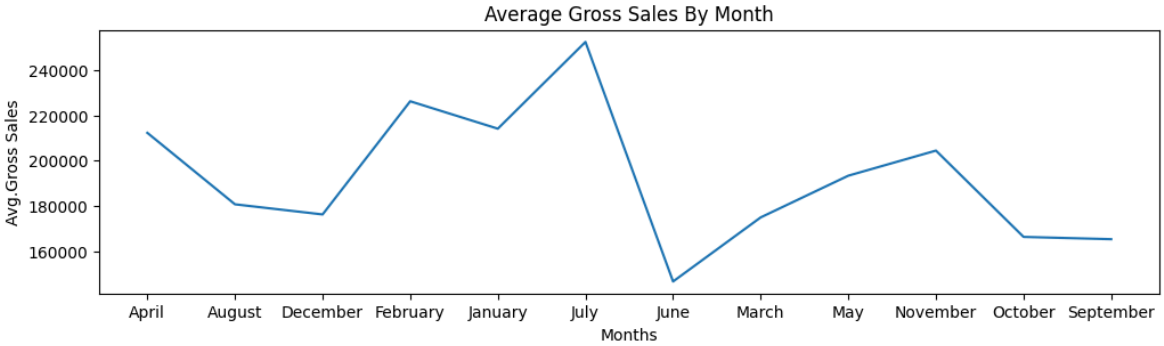
In [11]:
```
#2.Create a line chart showing monthly sales trend
```

In [14]:
```
#asbm=average gross sales by month
asbm=df.groupby('Month Name')['Gross Sales'].mean().reset_index()
asbm
```

Out[14]:

	Month Name	Gross Sales
0	April	212268.357143
1	August	180741.685714
2	December	176271.336538
3	February	226190.500000
4	January	214097.750000
5	July	252372.214286
6	June	146699.600000
7	March	174972.171429
8	May	193368.885714
9	November	204433.579710
10	October	166365.589928
11	September	165357.900000

In [21]:
```
plt.figure(figsize=(12,3))
plt.plot(asbm['Month Name'],asbm['Gross Sales'])
plt.title("Average Gross Sales By Month")
plt.xlabel("Months")
plt.ylabel("Avg.Gross Sales")
plt.show()
```



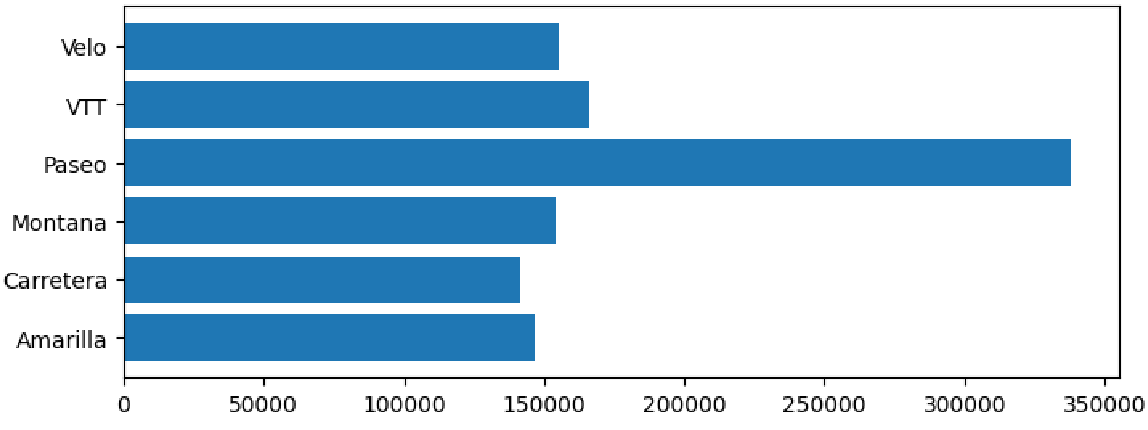
In [19]:
```
#3.Plot Units sold by product using a horizontal bar chart
```

In [23]:
```
#uspp is units sold per product
uspp=df.groupby('Product')['Units Sold'].sum().reset_index()
uspp
```

Out[23]:

	Product	Units Sold
0	Amarilla	146555.0
1	Carretera	141768.0
2	Montana	154198.0
3	Paseo	338239.5
4	VTT	165939.0
5	Velo	155143.5

In [24]:
```
plt.figure(figsize=(8,3))
plt.barh(uspp['Product'],uspp['Units Sold'])
plt.show()
```



In [25]:
```
#4.Create a histogram of profit distribution
```

In [26]:
```
pbm=df.groupby('Month Name')['Profit'].sum().reset_index()
pbm
```

Out[26]:

	Month Name	Profit
0	April	929984.57
1	August	791066.42
2	December	2717329.98
3	February	1148547.39
4	January	814028.68
5	July	923865.68
6	June	1473753.82
7	March	669866.87
8	May	828640.06
9	November	1370102.50
10	October	3439781.02
11	September	1786735.27

In [29]:

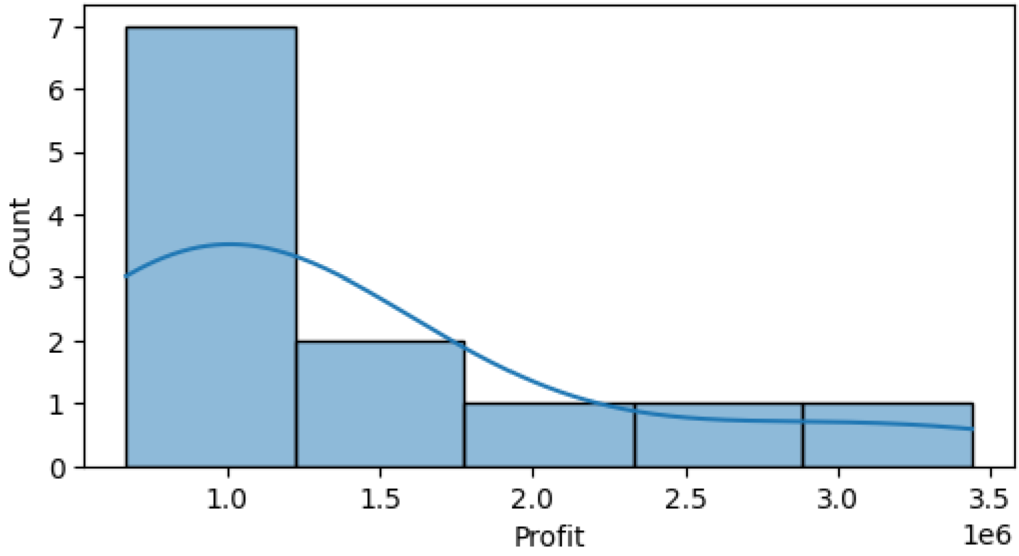
```
df.isnull().sum()
```

Out[29]:

```
Segment                 8
Country                11
Product                 8
Discount Band          62
Units Sold              5
Manufacturing Price     0
Sale Price              7
Gross Sales             5
Discounts               5
 Sales                  0
COGS                    0
Profit                  0
Date                    0
Month Number            0
Month Name              0
Year                    0
dtype: int64
```

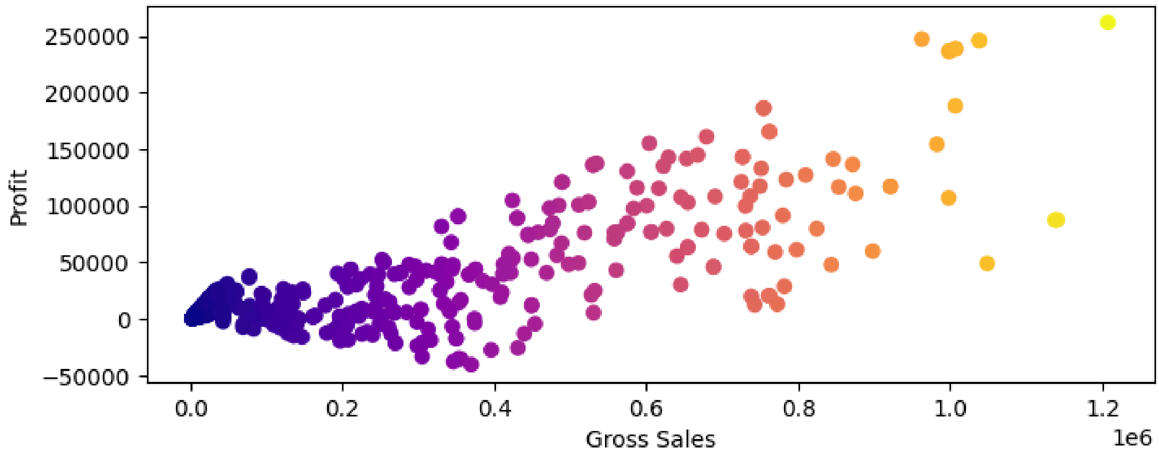
In [28]:

```
plt.figure(figsize=(6,3))
sns.histplot(pbm['Profit'],kde=True)
plt.show()
```



In [30]:
```
#5.plot sales vs profit using a scatter plot
```

In [41]:
```
x=df['Gross Sales']
y=df['Profit']
plt.figure(figsize=(8,3))
plt.scatter(x,y,c=x,cmap='plasma')
plt.xlabel("Gross Sales")
plt.ylabel("Profit")
plt.show()
```



In [42]:
```
#6.Create a pie chart showing profit contribution by segment
```

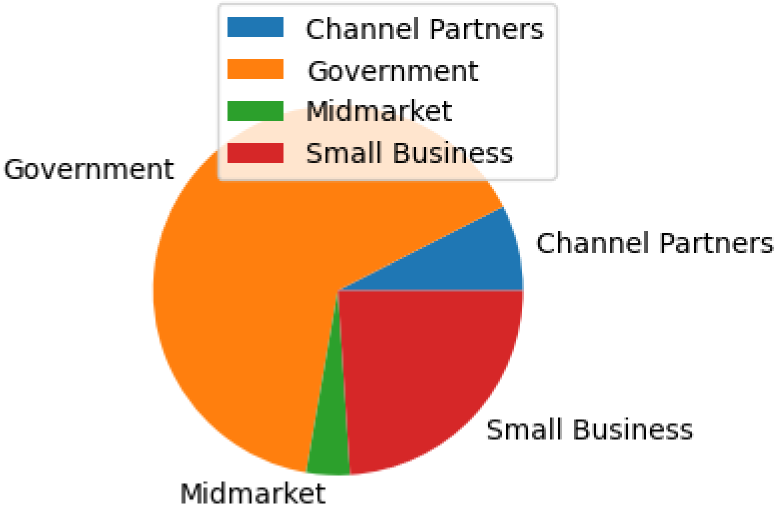
In [47]:
```
newdf=df.drop(df[df['Segment']=='Enterprise'].index)
```

In [48]:
```
pbs=newdf.groupby('Segment')['Profit'].sum().reset_index()
pbs
```

Out[48]:

	Segment	Profit
0	Channel Partners	1.290808e+06
1	Government	1.121272e+07
2	Midmarket	6.601031e+05
3	Small Business	4.143168e+06

In [72]:
```
plt.figure(figsize=(6,3))
plt.pie(pbs['Profit'],labels=pbs['Segment'])
plt.legend(
    labels=pbs['Segment'],
    loc="upper right",
    bbox_to_anchor=(1,1.15)
)
plt.show()
```



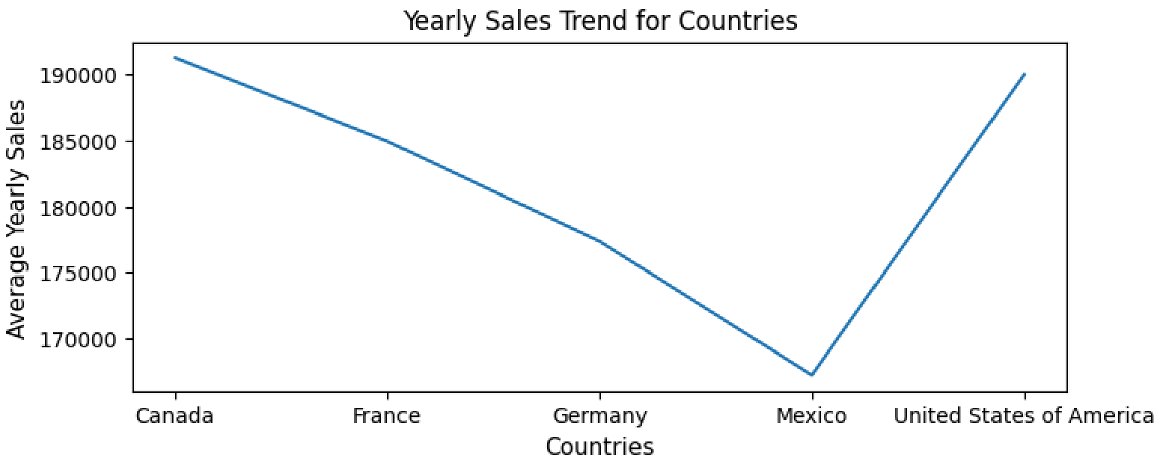
In [73]:
```
#7.Plot yearly sales trends for each country using a multi line chart
```

In [80]:
```
asbc=df.groupby('Country')['Gross Sales'].mean().reset_index()
asbc
```

Out[80]:

	Country	Gross Sales
0	Canada	191272.974265
1	France	184950.895683
2	Germany	177365.156934
3	Mexico	167222.651852
4	United States of America	190006.270073

In [93]:
```
plt.figure(figsize=(8,3))
plt.plot(asbc['Country'],asbc['Gross Sales'])
plt.title("Yearly Sales Trend for Countries")
plt.xlabel("Countries",fontsize=11)
plt.ylabel("Average Yearly Sales",fontsize=11)
plt.show()
```



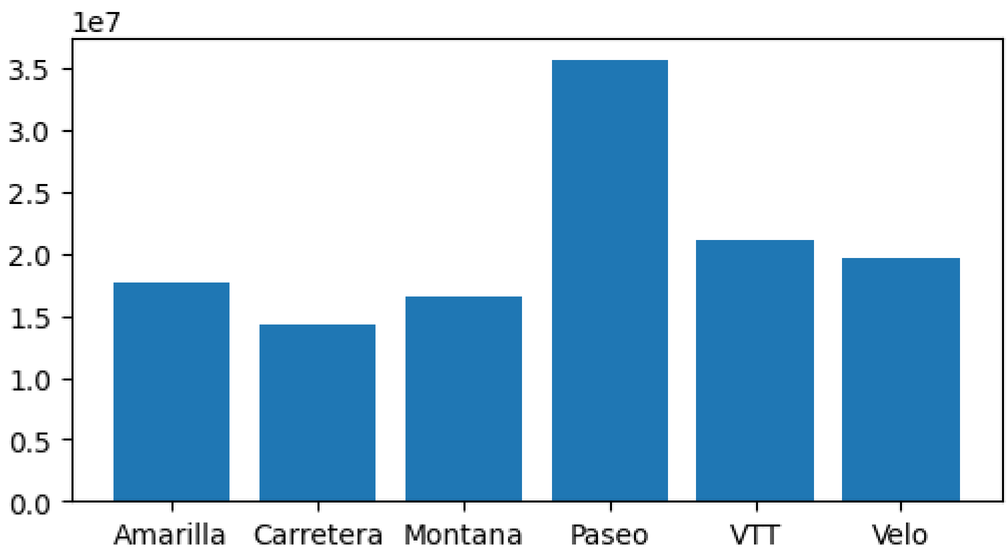
In [94]:
```
#8.Create a bar chart of gross sales by product
```

In [115…
```
gsbp=df.groupby('Product')['Gross Sales'].sum().reset_index()
gsbp
```

Out[115…

	Product	Gross Sales
0	Amarilla	17668368
1	Carretera	14326899
2	Montana	16526158
3	Paseo	35611661
4	VTT	21102442
5	Velo	19564046

In [123…
```
plt.figure(figsize=(6,3))
plt.bar(gsbp['Product'],gsbp['Gross Sales'])
plt.show()
```



In [124…
```
#9.Plot Total sales by segment using a bar chart
```

In [125…
```
gsbs=df.groupby('Segment')['Gross Sales'].sum().reset_index()
```

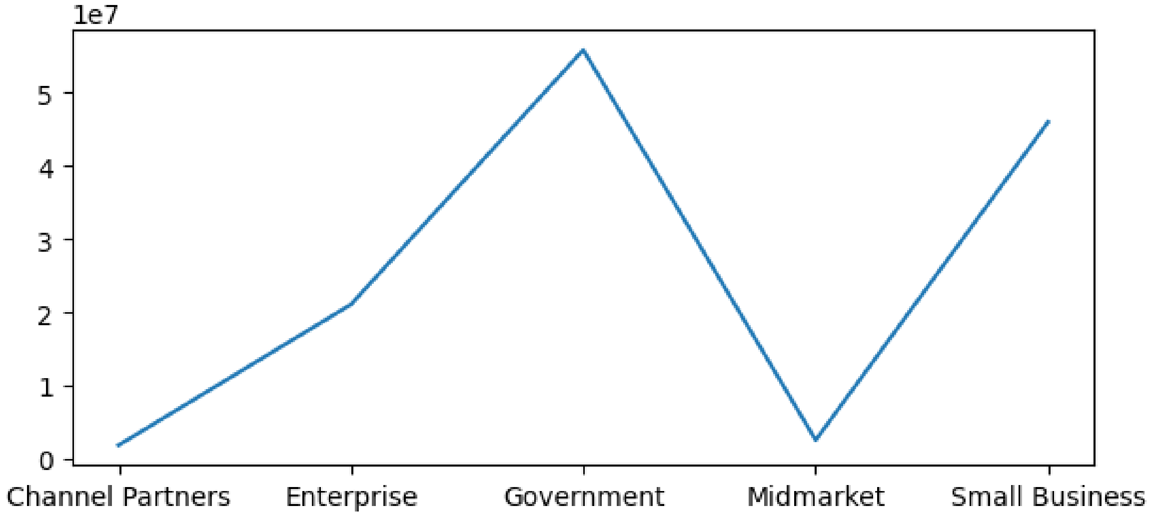
```
gsbs
```

Out[125…

	Segment	Gross Sales
0	Channel Partners	1871862
1	Enterprise	21068997
2	Government	55772475
3	Midmarket	2569777
4	Small Business	45941700

In [126…

```
plt.figure(figsize=(7,3))
plt.plot(gsbs['Segment'],gsbs['Gross Sales'])
plt.show()
```



In [127…

```
#10.Create a line chart of monthly profit over time
```

In [128…

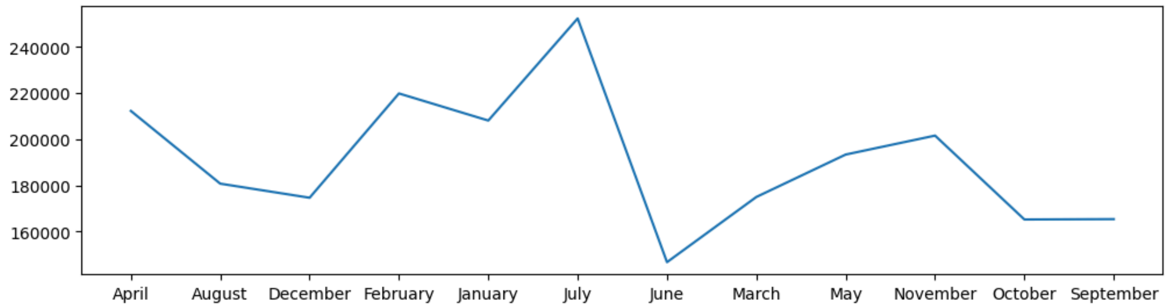
```
#asbm=average gross sales by month
asbm=df.groupby('Month Name')['Gross Sales'].mean().reset_index()
asbm
```

Out[128…

	Month Name	Gross Sales
0	April	212268.285714
1	August	180741.685714
2	December	174634.504762
3	February	219853.742857
4	January	208106.457143
5	July	252372.142857
6	June	146699.600000
7	March	174972.171429
8	May	193368.885714
9	November	201576.014286
10	October	165208.721429
11	September	165357.900000

In [130…

```
plt.figure(figsize=(12,3))
plt.plot(asbm['Month Name'],asbm['Gross Sales'])
plt.show()
```



In [ ]: